

Project Documentation: Moving Valmeyer Spatial Analysis

1. Objective

Following the catastrophic Mississippi River flooding of 1993, the town of Valmeyer, Illinois, was relocated to higher ground on nearby bluffs. The objective of this GIS project was to:

1. Perform spatial and zonal analysis on the relocated townsite (New Valmeyer) to assess terrain elevation, slope, and relocation distance from the original townsite (Old Valmeyer).
2. Integrate tabular statistics permanently into vector parcel attributes.
3. Digitize a complete road centerline network for the new town site.
4. Apply cartographic symbology and publish an interactive, annotated Web Map to ArcGIS Online for urban planning presentation.

2. Data

Vector Datasets

- **old_valmeyer_parcel**s: Polygon feature class representing parcel boundaries of the flood-prone historic townsite.
- **new_valmeyer_parcel**s: Polygon feature class representing parcel boundaries of the newly developed upland site.
- **New_Valmeyer_Roads**: Digitized polyline feature class representing the new road network centerlines.

Raster Datasets

- **valmeyer_dem_10m**: 10-meter resolution Digital Elevation Model (DEM) of the area.
- **Slope_valmey1**: Slope raster derived from the DEM, measured in degrees.
- **EucDist_old_1**: Euclidean distance raster measuring straight-line distance (in meters/feet) from the old town center.

Tabular Datasets

- **ZonalSt_new_val1**: Zonal statistics output table for Elevation (Min, Max, Mean).
- **ZonalSt_new_val2**: Zonal statistics output table for Slope (Min, Max, Mean).
- **ZonalSt_new_val3**: Zonal statistics output table for Euclidean Distance (Mean).

Basemaps

- **World Topographic Map / World Hillshade** (Esri).

3. Methodology

[DEM / Slope / Distance Rasters] —> (Zonal Statistics as Table) —> [Standalone Summary Tables]

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[new_valmeyer_parcel] <— (Remove Joins) <— (Calculate Field) <— (Table Join via PARCEL_NUM)

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| —> (Digitize Roads with Snapping) —> [New_Valmeyer_Roads]

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| —> (Graduated Colors Symbolology)

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| —> (Project to Web Mercator) —> (Publish to AGOL) —> [Interactive Web Map]

1. **Zonal Summary Tables:** Executed *Zonal Statistics as Table* for each raster layer against new_valmeyer_parcel using PARCEL_NUM as the unique zone field.
2. **Relational Joins & Attribute Transfer:**
 - Joined summary tables (ZonalSt_new_val1, val2, val3) to new_valmeyer_parcel via PARCEL_NUM.
 - Added 7 double-precision fields: min_elev, max_elev, mean_elev, min_slope, max_slope, mean_slope, and move_dist.
 - Used Python expressions in **Calculate Field** (e.g., !ZonalSt_new_val1.MIN!) to permanently populate values.
 - Removed all tabular joins once fields were fully populated.
3. **Vector Feature Digitizing:**
 - Created a Polyline feature class New_Valmeyer_Roads in the project File Geodatabase (Valmeyer_IL.gdb).
 - Configured Point, End, and Intersection vertex snapping.

- Digitized road centerlines along parcel gaps, producing a connected network.

4. **Symbology & Cartography:**

- Styled new_valmeyer_parcel using **Graduated Colors** based on mean_elev (Natural Breaks / Jenks, 5 classes).
- Styled old_valmeyer_parcel with a solid single symbol.
- Styled New_Valmeyer_Roads as a solid dark 2.0 pt line.

5. **Web Transformation & Publishing:**

- Transformed map coordinate system to **WGS 1984 Web Mercator (Auxiliary Sphere)**.
- Enabled unique layer IDs and published vector layers to ArcGIS Online.
- Added sketch layer map annotations labeling **Old Valmeyer** and **New Valmeyer**.

4. **Analysis**

- **Elevation Analysis:** new_valmeyer_parcel sit on an elevated bluff ranging approximately between 202m and 243m above sea level, placing the new community safely above Mississippi River flood stages.
- **Slope Suitability:** Most residential parcels exhibit gentle slopes (<15°), while peripheral parcels border steeper terrain (>20°), influencing development feasibility.
- **Relocation Distance:** Euclidean distance calculations confirm the new townsite relocated approximately 2,000m to 2,600m east of the historic flood zone.

5. **Result & Deliverables**

- **Final Web Map:** [Moving Valmeyer Final Web Map](#)
- **Local Source Package:** Moving Valmeyer.ppkx / Valmeyer_IL.gdb (containing updated vector layers and clean attribute data).
- **Validated Parcel Metrics:** All 480 parcel records permanently contain validated elevation, slope, and relocation distance attributes ready for spatial queries and reporting.

6. Screenshots:

